
VERIFICATION REPORT AND CERTIFICATION STATEMENT

United Carbon Assets Limited

**Inner Mongolia Wudaogou 50.25MW
Wind Power Project**

SGS Climate Change Programme

SGS United Kingdom Ltd
SGS House
217-221 London Road
Camberley Surrey
GU15 3EY
United Kingdom

Date of Issue:		Project Number:	
18-06-2009		CCP.VOL0291B	
Project Title:			
Inner Mongolia Wudaogou 50.25MW Wind Power Project			
Organisation:		Client:	
SGS United Kingdom Limited		United Carbon Assets Limited	
Publication of Monitoring Report:			
Monitoring Period:		01/01/2007 - 30/06/2008	
First Monitoring Report Version and Date:		Version 1, 25/09/2008	
Final Monitoring Report Version and Date:		Version 2, 08/05/2009	
Summary:			
SGS United Kingdom Ltd has performed the verification of Inner Mongolia Wudaogou 50.25MW Wind Power Project in accordance with VCS 2007.1, this project has been registered as a CDM project with the UNFCCC reference No. 1487 on 05/07/2008. The verification includes confirming the implementation of the monitoring plan of the registered CDM PDD Version 3.0 dated 14/11/2007 (also referred as "VCS PD" hereinafter) and the application of the monitoring methodology as per ACM0002 version 06. A site visit was conducted to verify the data submitted in the monitoring report. Inner Mongolia Wudaogou 50.25MW Wind Power Project is a grid connected renewable energy project located in Inner Mongolia Autonomous Region, Northeast China. The objective of the project is to generate electricity and to sell into the Northeast Power Grid of China (NEPG). The project achieves CO2 emission reduction by replacing electricity generated by fossil fuel fired power plants connected to NEPG. With an average annual electricity generation of 124.66GWh, the project is estimated to produce 142,848 tonnes of CO2 emission reductions annually. SGS confirms that the project is implemented in accordance with the registered CDM PDD. The monitoring system is in place and the emission reductions are calculated without material misstatements. Our opinion relates to the projects GHG emissions and the resulting GHG emission reductions reported and related to the valid and validated project baseline and monitoring and its associated documents. Based on the information seen and evaluated we confirm that the implementation of the project has resulted in 224,282 tCO2e Emission Reductions during period 01/01/2007 up to 30/06/2008.			
Subject:			
VCS 2007.1 Project Verification			
Verification Team:			
Simon Zhao Xinguang – Lead Assessor		☒ No Distribution (without permission from the Client or responsible organisational unit)	
Ginger Jiang Yuan – Assessor			
Technical Review:	Trainee Technical Reviewer:		☐ Limited Distribution
Date: 21-06-2009 and 28-06-2009 Name: Elton Chen Wu	Name: N/A		
Authorised Signatory:			☐ Unrestricted Distribution
Name: Siddharth Yadav			
Date: 13 th July 2009			
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Abbreviations

CAR	Corrective Action Request
CDM	Clean Development Mechanism
CEF	Carbon Emission Factor
CL	Clarification Request
CO ₂	Carbon Dioxide
CO ₂ e	Carbon Dioxide Equivalent
DNA	Designated National Authority
DOE	Designated Operational Entity
DRRs	Daily Reading Records
GHG	Greenhouse Gas(es)
GWP	Global Warming Potential
IPCC	Intergovernmental Panel on Climate Change
MP	Monitoring Plan
MR	Monitoring Report
MRRs	Monthly Reading Records
NEPG	Northeast Power Grid
ODA	Official Development Assistance
PP	Project Participant
PPA	Power Purchase Agreement
SGS	Société Générale de Surveillance
UNFCCC	United Nations Framework Convention on Climate Change
VCS	Voluntary Carbon Standard
VCS PD	VCS Project Description
VCU	Voluntary Carbon Unit

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1. Introduction

1.1 Objective

SGS United Kingdom Ltd has been contracted by United Carbon Asset Limited to perform an independent verification of its VCS project Inner Mongolia Wudaogou 50.25MW Wind Power Project. The verifiers have reviewed the GHG data collected to date for the period between 01/01/2007 and 30/06/2008

The purposes of this verification exercise are, by review of objective evidence, to independently review:

- Whether the project has resulted in emission reductions as declared by the organisation or GHG project's GHG assertion;
- The data reported are accurate, complete, consistent, transparent and free of material error or omission.

1.2 Scope and Criteria

This engagement covers verification of emission reductions from anthropogenic sources of greenhouse gases included within the project boundary of the Inner Mongolia Wudaogou 50.25MW Wind Power Project.

Our approach is risk-based, drawing on an understanding of the risks associated with reporting GHG emissions data and the controls in place to mitigate these. Our examination includes assessment, on a test basis, of evidence relevant to the amounts and disclosures in relation to the project's GHG emission reductions for the defined reporting period.

The verification is not meant to provide any consulting towards the Client. However, stated requests for clarifications and/or corrective actions may provide input for improvement of the project design.

1.3 VCS project Description

Inner Mongolia Wudaogou 50.25MW Wind Power Project is a grid connected renewable energy project located in Inner Mongolia Autonomous Region, Northeast China. The objective of the project is to generate electricity and to sell into the Northeast Power Grid of China (NEPG). The project achieves CO₂ emission reduction by replacing electricity generated by fossil fuel fired power plants connected to NEPG. With an average annual electricity generation of 124.66GWh, the project is estimated to produce 142,848 tonnes of CO₂ emission reductions annually.

1.4 Level of assurance

The level of assurance of the verification report is reasonable assurance engagements as selected by the Client. Materiality for the project is 5%.

1.5 Project Activity and Period Covered

This engagement covers emissions and emission reductions from anthropogenic sources of greenhouse gases included within the project boundary of the following project and period.

Title of Project Activity:	Inner Mongolia Wudaogou 50.25MW Wind Power Project
SGS Reference No:	CCP.VOL0291B
Monitoring Period Covered in this Report	01/01/2007 - 30/06/2008
Project Owner	Chifeng Xinsheng Wind Power Co. Ltd (People's Republic of China)
Location of the Project Activity:	Wengniute County, Inner Mongolia, P.R.China

2. Methodology

2.1 General Approach

SGS's approach to the verification is a two-stage process.

In the first stage, SGS completed a strategic review and risk assessment of the projects activities and processes in order to gain a full understanding of:

- Activities associated with all the sources contributing to the project emissions and emission reductions, including leakage if relevant;
- Protocols used to estimate or measure GHG emissions from these sources;
- Collection and handling of data;
- Controls on the collection and handling of data;
- Means of verifying reported data; and
- Compilation of the monitoring report.

At the end of this stage, SGS produced a Verification Checklist which, based on the risk assessment of the parameters and data collection and handling processes for each of those parameters, describes the verification approach and the sampling plan.

Using the Verification checklist, SGS verified the implementation of the monitoring plan and the data presented in the Monitoring Report for the period in question. This involved a site visit and a desk review of the monitoring report. This verification report describes the findings of this assessment.

2.2 Verification Team for this Assessment

Name	Role	SGS Office
Simon Zhao Xinguang	Lead Assessor	SGS China
Ginger Jiang Yuan	Assessor	SGS China

2.3 Means of Verification

2.3.1 Review of Documentation

The registered CDM PDD, serving as the VCS PD, the monitoring report submitted by the client and additional background documents related to the project performance were reviewed. A complete list of all documents reviewed is attached in section 7 of this report.

2.3.2 Site Visits

As part of the verification, the following on-site inspections have been performed

Location: Wengniute County, Inner Mongolia, P.R.China	
Date: 04/11/2008-05/11/2008	
Coverage:	Source of Information / Persons Interviewed
Review of the monitoring report, metering reading records, invoices and monthly reading records.	Mr. Cui Baohe, Chifeng Xinsheng Wind Power Co. Ltd;
An interview with project participants including reviewing the installation plan and key physical components of the project, VCS monitoring report and staff training records.	Mr. Wang Tao, United Carbon Assets Limited; Mr. Wang Zhenning, United Carbon Assets Limited.
Collection of the calibration certificates for all the meters involved and the accreditation certificate of the calibration entity.	
Inspection of the control room, instrument room of the project.	

2.4 Reporting of Findings

As an outcome of the verification process, the team can raise different types of findings

In general, where insufficient or inaccurate information is available and clarification or new information is required the team shall raise a Clarification Request (CL) specifying what additional information is required.

Where a non-conformance arises the team shall raise a Corrective Action Request (CAR). A CAR is issued, where:

- I. the verification is not able to obtain sufficient evidence for the reported emission reductions or part of the reported emission reductions. In this case these emission reductions shall not be verified and certified;
- II. the verification has identified misstatements in the reported emission reductions. Emission reductions with misstatements shall be discounted based on the verifiers ex-post determination of the achieved emission reductions

The verification process may be halted until this information has been made available to the assessors' satisfaction. Failure to address a CL may result in a CAR. Information or clarifications provided as a result of a CL may also lead to a CAR.

FARs may be raised which are for the benefit of future projects and future verification actors. These have no impact upon the completion of the verification activity.

Corrective Action Requests and Clarification Requests are detailed in Verification Checklist. The Project Developer is given the opportunity to "close" outstanding CARs and respond to CLs and FARs.

2.5 Internal Quality Control

Following the completion of the assessment process and a recommendation by the Assessment Team, all documentation will be forwarded to a Technical Reviewer. The task of the Technical Reviewer is to check that all procedures have been followed and all conclusions are justified. The Technical Reviewer will either accept or reject the recommendation made by the assessment team.

3. Verification Findings

3.1 Project Documentation and Compliance with the registered PDD and VCS PD

3.1.1 Validation opinion of clauses 1.12, 1.13, 1.14, 8.1 and 8.2 of the VCS Project Description Template

According to Section 5.2.1 of Voluntary Carbon Standard 2007.1 dated 18/11/2008, for non-AFOLU projects, VCS 2007.1 validation shall be completed within two years of the project start date, or shall be completed or contracted before 19 November 2008. United Carbon Asset Limited contracted SGS for the verification of the project on 10/10/2008, which is before 19/11/2008.

Pursuant to section 4.1.4 of VCS Guidance Document: VCS Project Registration and VCU Issuance Process version 1.1 (/15/), a further validation has been completed of clauses 1.12, 1.13, 1.14, 8.1 and 8.2 of the VCS Project Description Template. Such further validation is included in this verification report and details are presented as below:

Clause 1.12: Demonstration to confirm that the project was not implemented to create GHG emissions primarily for the purpose of its subsequent removal or destruction.

Through onsite visit, it is validated that the project is an energy generation project with renewable wind power. The GHG emissions from the project are considered to be 0. It was not implemented to create GHG emissions primarily for the purpose of its subsequent removal or destruction.

Clause 1.13: Demonstration that the project has not created another form of environmental credit (for example renewable energy certificates).

Through interview with the project owner and document review of the statement from the project owner confirming that the project doesn't create another form of environmental credit, it is validated that the project has not created another form of environmental credit. The project owner provided a statement indicating that the project has not created another form of environmental credit (/12/).

Clause 1.14: Project rejected under other GHG programs (if applicable).

Through interview with the project owner and review of the UNFCCC website, it is validated that the project has been registered as a CDM project on 05/07/2008 and has not been rejected under other GHG programs.

Clause 8.1: Proof of Title.

The Chinese DNA issued the LoA (/13/) for the project to participate in CDM on 13/12/2006. In the LoA, it is indicated that Chifeng Xinsheng Wind Power Co. Ltd is authorized as China's participant to voluntarily participate in and carry out the project activity. The Chinese DNA had examined the ownership of the project before it issued the LoA to the project participant. The LoA from the Chinese DNA is the proof of title.

Clause 8.2: Projects that reduce GHG emissions from activities that participate in an emissions trading program (if applicable).

Through interview with the project owner, it is validated that the project owner is not included in another voluntary emission trading programme listed in the website of VCS (www.v-c-s.org) during this monitoring period. The project owner has provided a statement and further confirmed in the monitoring report version 2 that the ERs generated from the project in this monitoring period will be sold only once.

The project takes place in China, which has no emission reductions target and the project is not a micro or mega project. There are no mandatory emission trading programmes in China. The project is not a grouped one and the project is a registered CDM project activity and not rejected under other GHG programs. The PP has confirmed that there is no financially sensitive information that is provided to the DOE.

The project helps improve local economy, increase local financial income and promote electricity utilization in the local area. The project is in compliance with all the relevant local laws and regulations for environmental protection, energy conservation and sustainable development. The project has obtained approvals from relevant authorities. There are no risks that may substantially affect the project's GHG emission reductions.

As per the validation report for the project for CDM registration and the above further validation of Clauses 1.12, 1.13, 1.14, 8.1 and 8.2 of the VCS PD, it is validated that the Project “Inner Mongolia Wudaogou 50.25MW Wind Power Project” is in compliance with the VCS Program requirements as set out in the VCSA Rules.

3.1.2 Compliance with the registered PDD and VCS PD

The project is a GHG emission reductions project. The project has been registered as a CDM project activity against ACM0002 version 06 (/1/). The registration date for the project is 05/07/2008(/5/). Requirements of VCS 2007.1 are fulfilled in the Monitoring Report version 2 dated 08/05/2009 (/3/).

This is the first verification for the emission reductions achieved before the start of CDM crediting period for the project under VCS2007.1. The monitoring period covered in this verification is from 01/01/2007 to 30/06/2008. The end date of this monitoring period is before the start date of the CDM crediting period for the project. The project has not passed a ten year crediting period. According to 5.2.1 of Voluntary Carbon Standard 2007.1 dated 18/11/2008, crediting period start date for any non-AFOLU project under the VCS 2007.1 shall not be earlier than 28/03/2006; the starting date of this monitoring period of this project is 01/01/2007.

The monitoring report is consistent with ACM0002 version 06 against which the project was validated and registered. There are no differences between ACM0002 version 06 and the monitoring plan in the CDM PDD V3.0 dated 14/11/2007(/2/). The project boundary is consistent with the registered CDM PDD.

As the project shares substation and transmission lines with another wind farm project (Inner Mongolia Sunjiaying 50.25MW Wind Power Project, which belongs to the same project owner with the project and has been registered as a CDM project with the registration number 1488), the original monitoring plan in the registered CDM PDD is not able to monitor all parameters that are need to calculated Emission Reductions. The monitoring plan is revised. The main changes to the original monitoring plan are as follows:

Parameters monitored are:

E_{Export} : Total electricity delivered to the grid by Wudaogou and Sunjiaying projects;

E_{Import} : Total electricity imported from the grid by Wudaogou and Sunjiaying projects;

E_{Wudaogou} : Electricity generated by Wudaogou project;

$E_{\text{Sunjiaying}}$: Electricity generated by Sunjiaying project;

$E_{10\text{kV}}$: Electricity imported from 10kv backup line;

And EG_y is calculated as follows:

$$EG_y = E_{\text{Export}} / (E_{\text{Sunjiaying}} + E_{\text{Wudaogou}}) * E_{\text{Wudaogou}} - E_{\text{Import}} - E_{10\text{kV}}$$

Description of measurement methods and procedures to be applied and QA/QC procedures to be applied are specified for each parameter in the revised monitoring plan, which is attached in Annex 4 to the verification report. The meter location and accuracy class of each meter and the calibration procedures are specified in the revised monitoring plan.

Validation Opinion for the Revised Monitoring Plan:

All parameters that are necessary for the calculation of emission reductions are included in the revised monitoring plan. The calculation method of EG_y is conservative.

It is confirmed by SGS that:

- (1) The revised monitoring plan is in accordance with the methodology applied by the project, i.e. ACM0002, version 06;
- (2) The revision of the monitoring plan ensures that the level of accuracy or completeness in the monitoring and verification process is not reduced as a result of the revision;
- (3) This is the first verification of the project and there are no findings from previous verification report need to be taken into account.

3.2 Monitoring Results

As the project shares transmission lines and substation with another wind farm project, Inner Mongolia Sunjiaying 50.25MW Wind Power Project, which is also a registered CDM project with the UNFCCC reg. No. 1488, E_{G_y} is calculated as follows:

$$E_{G_y} = E_{\text{Export}} / (E_{\text{Sunjiaying}} + E_{\text{Wudaogou}}) * E_{\text{Wudaogou}} - E_{\text{Import}} - E_{10kv}$$

Where,

E_{Export} is the total electricity supplied to the grid, this data is provided by the grid company and double checked with sales receipts and it is checked by the meter reading from the onsite substation (which is for the two wind farms combined);

$E_{\text{Sunjiaying}}$ is the electricity generation metered by the onsite meters of 1488 Inner Mongolia Sunjiaying 50.25MW Wind Power Project (SGS is also contracted to take the verification of this project against VCS 2007.1);

E_{Wudaogou} is the electricity generation metered by the onsite meters of 1487 Inner Mongolia Wudaogou 50.25MW Wind Power Project;

E_{Import} is the total electricity imported from the grid by the two projects; this data is measured by the grid company and confirmed by electricity purchase invoice;

E_{10kv} is the electricity consumption through the 10kV back-up power line by the project activity, which is also confirmed by the grid company.

All the meters above were read and recorded by designated personnel from the wind farm company at 24:00 every day. And the daily readings were transcribed into Daily Reading Records (DRRs, /8/) and the readings at 24:00 on the last day of every month were transcribed into Monthly Reading Records (MRRs, /9/). The meter measuring E_{Export} and E_{Import} (gateway meter) was also read and recorded by designated personnel from the grid company on the last day of every month. The grid company informed the wind farm company about the amount of electricity exported to the grid by the 2 projects and invoices were issued by the wind farm company based on the conformed amount of exported electricity. Data in MRRs were used for emission reduction calculations and sales receipts were used for double checking. Conservative values were chosen.

In version 1 of the monitoring report, the calibration information of the meters was not reported. **CL #1 was raised** for requesting the PP to clarify how the meters were calibrated as per the registered CDM PDD and to provide related evidences. The PP reported the meter information and calibration information in version 2 of the monitoring report and provided the calibration records of all the meters involved and the accreditation certificates of the calibrating entities. The calibration records and accreditation certificates of the calibrating entities were verified by the assessment team. It is confirmed that the meters were calibrated annually as per the registered CDM PDD. **CL #1 is closed out.** The meter information and calibration information is as follows:

Data measured	Manufacturer	Type	Serial number	Accuracy	Calibration date
$E_{\text{Export}} \& E_{\text{Import}}$	Landisgyr	ZD402	86693424	0.2S	25/10/2007 09/11/2006
E_{Wudaogou}	Changsha Weisheng	DTSD341	200504230U0190	0.5S	19/05/2008 19/05/2007 16/11/2006
	Changsha Weisheng	DTSD341	200504230U0242	0.5S	19/05/2008 19/05/2007 16/11/2006
$E_{\text{Sunjiaying}}$	Changsha Weisheng	DTSD341	200504230U0186	0.5S	19/05/2008 19/05/2007 16/11/2006
	Changsha Weisheng	DTSD341	200504230U0245	0.5S	19/05/2008 19/05/2007 16/11/2006
E_{10kv}	Zhejiang Zhengtai	DTS634	2006-04-46355251	1.0S	19/05/2008 28/05/2007 28/05/2006

The calculation of EG_y and emission reductions was carried out by using the ER spreadsheet (/10/). The ER spreadsheet was verified. The values used in the spreadsheet and the calculation process were verified to be correct. It was confirmed that the calculation of emission reductions is correct in the ER spreadsheet. And the calculation results were correctly reported in version 2 of the monitoring report.

3.3 Remaining Issues, CAR's, CL's from Previous Validation or Verification

This is the first verification for the project against VCS 2007.1. There are no remaining issues from previous validation process.

3.4 Project Implementation

Project was implemented and equipments were installed as described in the registered PDD.

3.5 Completeness of Monitoring

The reporting procedures reflect the content of the monitoring plan. The monitoring mechanism is effective and reliable.

3.6 Accuracy of Emission Reduction Calculations

The calculation of emission reductions is found to be correct. No findings regarding accuracy of emission reduction calculations were raised. The details of the reported and the verified values for all parameters are listed in section 4.

3.7 Quality of Evidence to Determine Emission Reductions

Critical parameters used for the determination of the Emission Reductions are discussed above in section 3.2 above. All the data recorded is in compliance with the monitoring report.

3.8 Management and Operational System and Quality Assurance

Management system and quality assurance procedures have been stipulated in the monitoring plan and have been implemented during daily operation (/11/). This has been verified by on-site visit and document review. Therefore we can affirm that the management system of the VCS project is in place; with the responsibilities properly identified and in place.

3.9 Data from External Sources

Baseline emission factor is calculated from external data. It was determined ex-ante and fixed for this crediting period according to description in the registered CDM PDD and the validation report. The value of the baseline emission factor used in the monitoring report version 2 is 1.1459tCO₂e/MWh. It complies with the calculated value of the baseline emission factor in the registered CDM PDD.

4. Calculation of Emission Reductions

Table 1 Reported and Verified Data of E_{Export}

Period	Reported Value (kWh)	Verified Value(kWh)
01/01/07-31/01/07	23,700,000	23,700,000
01/02/07-28/02/07	28,640,000	28,640,000
01/03/07-31/03/07	19,040,000	19,040,000
01/04/07-30/04/07	22,820,000	22,820,000
01/05/07-31/05/07	24,346,740	24,346,740
01/06/07-30/06/07	6,420,000	6,420,000
01/07/07-31/07/07	4,490,000	4,490,000
01/08/07-31/08/07	9,301,600	9,301,600
01/09/07-30/09/07	6,040,000	6,040,000
01/10/07-31/10/07	26,640,000	26,640,000
01/11/07-30/11/07	41,250,000	41,250,000
01/12/07-31/12/07	46,161,940	46,161,940
01/01/08-31/01/08	45,500,000	45,500,000
01/02/08-29/02/08	48,460,000	48,460,000
01/03/08-31/03/08	37,510,000	37,510,000
01/04/08-30/04/08	37,490,000	37,490,000
01/05/08-31/05/08	35,751,980	35,751,980
01/06/08-30/06/08	9,035,400	9,035,400
Total	472,597,660	472,597,660

Table 2 Reported and Verified Data of E_{Import}

Period	Reported Value (kWh)	Verified Value(kWh)
01/01/07-31/01/07	94,380	94,380
01/02/07-28/02/07	26,180	26,180
01/03/07-31/03/07	125,840	125,840
01/04/07-30/04/07	43,340	43,340
01/05/07-31/05/07	31,900	31,900
01/06/07-30/06/07	91,960	91,960
01/07/07-31/07/07	124,960	124,960
01/08/07-31/08/07	109,340	109,340
01/09/07-30/09/07	105,380	105,380
01/10/07-31/10/07	96,800	96,800
01/11/07-30/11/07	65,340	65,340
01/12/07-31/12/07	32,340	32,340
01/01/08-31/01/08	73,480	73,480
01/02/08-29/02/08	44,440	44,440
01/03/08-31/03/08	62,040	62,040
01/04/08-30/04/08	67,100	67,100
01/05/08-31/05/08	66,440	66,440
01/06/08-30/06/08	121,220	121,220
Total	1,382,480	1,382,480

Table 3 Reported and Verified Data of E_{Sunjiaying}

Period	Reported Value (kWh)	Verified Value(kWh)
01/01/07-31/01/07	11,398,800	11,398,800
01/02/07-28/02/07	14,074,550	14,074,550
01/03/07-31/03/07	9,941,750	9,941,750
01/04/07-30/04/07	11,558,750	11,558,750
01/05/07-31/05/07	7,487,550	7,487,550
01/06/07-30/06/07	3,330,600	3,330,600
01/07/07-31/07/07	4,807,950	4,807,950
01/08/07-31/08/07	2,065,700	2,065,700
01/09/07-30/09/07	3,761,450	3,761,450
01/10/07-31/10/07	3,138,450	3,138,450
01/11/07-30/11/07	14,788,200	14,788,200
01/12/07-31/12/07	26,036,150	26,036,150
01/01/08-31/01/08	28,031,150	28,031,150
01/02/08-29/02/08	29,636,950	29,636,950
01/03/08-31/03/08	30,705,850	30,705,850
01/04/08-30/04/08	28,035,000	28,035,000
01/05/08-31/05/08	25,352,950	25,352,950
01/06/08-30/06/08	23,843,400	23,843,400
Total	5,711,650	5,711,650
	283,706,850	283,706,850

Table 4 Reported and Verified Data of E_{Wudaogou}

Period	Reported Value (kWh)	Verified Value(kWh)
01/01/07-31/01/07	12,094,250	12,094,250
01/02/07-28/02/07	15,089,900	15,089,900
01/03/07-31/03/07	10,912,300	10,912,300
01/04/07-30/04/07	10,221,750	10,221,750
01/05/07-31/05/07	6,488,300	6,488,300
01/06/07-30/06/07	2,881,900	2,881,900
01/07/07-31/07/07	4,927,300	4,927,300
01/08/07-31/08/07	2,354,450	2,354,450
01/09/07-30/09/07	2,727,200	2,727,200
01/10/07-31/10/07	3,177,650	3,177,650
01/11/07-30/11/07	12,820,500	12,820,500
01/12/07-31/12/07	16,751,000	16,751,000
01/01/08-31/01/08	18,452,350	18,452,350
01/02/08-29/02/08	19,304,250	19,304,250
01/03/08-31/03/08	19,533,150	19,533,150
01/04/08-30/04/08	16,051,700	16,051,700
01/05/08-31/05/08	13,493,200	13,493,200
01/06/08-30/06/08	11,959,850	11,959,850
Total	3,442,600	3,442,600
	202,683,600	202,683,600

Table 5 Reported and Verified Data of E_{10kv}

Period	Reported Value (kWh)	Verified Value(kWh)
01/01/07-31/01/07	4,860	4,860
01/02/07-28/02/07	0	0
01/03/07-31/03/07	0	0
01/04/07-30/04/07	0	0
01/05/07-31/05/07	60	60
01/06/07-30/06/07	0	0
01/07/07-31/07/07	600	600
01/08/07-31/08/07	60	60
01/09/07-30/09/07	16,440	16,440
01/10/07-31/10/07	14,040	14,040
01/11/07-30/11/07	6,960	6,960
01/12/07-31/12/07	9,540	9,540
01/01/08-31/01/08	5,520	5,520
01/02/08-29/02/08	3,840	3,840
01/03/08-31/03/08	4,600	4,600
01/04/08-30/04/08	31,540	31,540
01/05/08-31/05/08	18,300	18,300
01/06/08-30/06/08	9,720	9,720
Total	126,080	126,080

Table 6 Reported and Verified Data of EG_y

Period	Reported Value (kWh)	Verified Value(kWh)
01/01/07-31/01/07	12,101,548	12,101,548
01/02/07-28/02/07	14,792,366	14,792,366
01/03/07-31/03/07	9,837,222	9,837,222
01/04/07-30/04/07	10,666,255	10,666,255
01/05/07-31/05/07	11,268,311	11,268,311
01/06/07-30/06/07	3,157,393	3,157,393
01/07/07-31/07/07	2,266,097	2,266,097
01/08/07-31/08/07	3,800,092	3,800,092
01/09/07-30/09/07	2,916,923	2,916,923
01/10/07-31/10/07	12,259,830	12,259,830
01/11/07-30/11/07	16,076,912	16,076,912
01/12/07-31/12/07	18,282,822	18,282,822
Subtotal of year 2007	117,425,770	117,425,770
01/01/08-31/01/08	17,867,911	17,867,911
01/02/08-29/02/08	18,793,187	18,793,187
01/03/08-31/03/08	13,590,523	13,590,523
01/04/08-30/04/08	12,923,502	12,923,502
01/05/08-31/05/08	11,857,984	11,857,984
01/06/08-30/06/08	3,266,965	3,266,965
Subtotal of year 2008	78,300,071	78,300,071
Total	195,725,841	195,725,841

The net quantity of electricity supplied to the grid by the project (EG_y) during the monitoring period from 01/01/2007 to 30/06/2008 is 195,725.841 MWh;

The baseline emission factor (EF_y) is 1.1459 tCO₂e/MWh;

Baseline emissions (BE_y) for the monitoring period from 01/01/2007 to 30/06/2008 is
 $BE_y = EG_y \times EF_y = 195,725.841 \text{ MWh} \times 1.1459 \text{ tCO}_2\text{e/MWh} = 224,282 \text{ tCO}_2\text{e}$;

Project emissions (PE_y) is 0;

Leakage (L_y) is 0;

According to the registered PDD, Emission Reductions for the monitoring period from 01/01/2007 to 30/06/2008 (ER_y) is calculated to be:

$$ER_y = BE_y - PE_y - L_y = 224,282 \text{ tCO}_2\text{e} - 0 - 0 = 224,282 \text{ tCO}_2\text{e}.$$

For the monitoring period in Year 2007:

Baseline emissions (BE_y) for the monitoring period from 01/01/2007 to 31/12/2007 is $BE_y = EG_y \times EF_y = 117,425.770 \text{ MWh} \times 1.1459 \text{ tCO}_2\text{e/MWh} = 134,558 \text{ tCO}_2\text{e}$;

Project Emissions (PE_y) is 0;

Leakage (L_y) is 0;

$$ER_y = BE_y - PE_y - L_y = 134,558 \text{ tCO}_2\text{e} - 0 - 0 = 134,558 \text{ tCO}_2\text{e}.$$

For the monitoring period in Year 2008:

Baseline emissions (BE_y) for the monitoring period from 01/01/2007 to 30/06/2008 is $BE_y = EG_y \times EF_y = 78,300.071 \text{ MWh} \times 1.1459 \text{ tCO}_2\text{e/MWh} = 89,724 \text{ tCO}_2\text{e}$;

Project Emissions (PE_y) is 0;

Leakage (L_y) is 0;

$$ER_y = BE_y - PE_y - L_y = 89,724 \text{ tCO}_2\text{e} - 0 - 0 = 89,724 \text{ tCO}_2\text{e}.$$

5. Recommendations for Changes in the Monitoring Plan

A revision of monitoring plan was made during this verification.

6. VCU Certification Statement

The scope of the verification

We, SGS have been engaged by United Carbon Assets Limited to certify that the greenhouse gas (GHG) emission reductions reported for Inner Mongolia Wudaogou 50.25MW Wind Power Project for the period from 01/01/2007 to 30/06/2008 in the verification report and a monitoring report version 2 dated 08/05/2009 are eligible for registration as Voluntary Carbon Units.

This engagement covers the verification of emission reductions from anthropogenic sources of greenhouse gases included within the project boundary of the Inner Mongolia Wudaogou 50.25MW Wind Power Project, as well as an additional confirmation of the compliance of the registered PDD with the requirements of VCS 2007.1.

Our approach is risk-based, drawing on an understanding of the risks associated with reporting GHG emissions data and the controls in place to mitigate these. Our examination includes assessment, on a test basis, of evidence relevant to the amounts and disclosures in relation to the project's GHG emission reductions for the defined reporting period.

The verification is not meant to provide any consulting towards the Client. However, stated requests for clarifications and/or corrective actions may provide input for improvement of the project design.

Conclusions of the verification

(a) We are an entity accredited by the United Nations Framework Convention for Climate Change (UNFCCC) to undertake certification and verification services in the sector in which the Project is undertaken. The accreditation is accepted by VCS Board as indicated in Clause 9 of VCS 2007.1.

(b) The monitoring report, together with other information examined, was prepared in full substance and form as required by the Voluntary Carbon Standard version 2007.1, published by the Climate Group as approved by the Voluntary Carbon Standard Steering Committee on 18/11/2008.

(c) The information in the Project's monitoring report together with other information examined by us, including all the information necessary for us to determine that the emission reductions achieved have been determined correctly.

(d) Based on our examination of the monitoring report and other relevant information, the Project meets all the requirements of the Voluntary Carbon Standard verification criteria contained in the VCS 2007.1.

(e) Based on our examination of the monitoring report and other relevant information, the claiming emission reductions during the monitoring period from 01/01/2007 to 30/06/2008 are verified as 224,282 tonnes CO₂ equivalent.

Liability statement with regards to the accuracy of the verification statement

The management of the project is responsible for the preparation of the GHG emissions data and the reported GHG emissions reductions. SGS is responsible for verification and confirming emission estimates for the project, as described in the monitoring report.

Our certification approach draws on an understanding of the risks associated with reporting GHG emissions data and the controls in place to mitigate these. Our examination includes an assessment of evidence, through desk review, and where necessary, interviews, stakeholder discussions and site visits, relevant to certifying the rightfulness of the amounts and disclosures in relation to the Project's GHG emission reductions.

We planned and performed our work to obtain the information and explanations that we considered necessary to provide sufficient evidence for us to give reasonable assurance that the amount of GHG emission reductions for the given period, prepared on the basis of the monitoring report, are fairly stated.

This assessment included:

- Collection of evidence supporting the reported data

- Checking whether the provisions of the Monitoring Plan in the VCS PD, were consistently and appropriately applied

We have verified whether the information included in the monitoring report representing the emission reductions achieved has been determined correctly for the given period from the baseline figure.

Certification statements

Based on process and procedures conducted, in our opinion, the monitoring report Version 2 dated 08/05/2009 on emission reductions for Inner Mongolia Wudaogou 50.25MW Wind Power Project during the reporting period from 01/01/2007 to 30/06/2008 is materially correct and is a fair representation of the GHG data and information and the emission reductions are fairly stated. All relevant facts have been found correct by our examination. The GHG emission reductions were calculated correctly on the basis of ACM0002 Version 06.

Therefore, SGS is able to certify that the project is in full compliance with the Voluntary Carbon Standard 2007.1, and the quantity of the reported emission reductions during below reporting period are completely, comparably, accurately and correctly reported.

Reporting period: From 01/01/2007 to 30/06/2008

Verified emission in the above reporting period:

Project emissions **0** t CO₂ equivalents

Baseline emissions **224,282** t CO₂ equivalents

Emission reductions **224,282** t CO₂ equivalents

Wherein

Unit: tCO₂ equivalents

Reporting Period	01/01/2007 to 31/12/2007	01/01/2008 to 30/06/2008
Project emissions	0	0
Baseline emissions	134,558	89,724
Emission Reductions	134,558	89,724

Statement of Confidentiality

SGS will hold all information confidential until the client instructs otherwise or until it has been released in accordance with the VCS 2007.1 modalities and procedures.

Signed on behalf of the Verification Body by Authorized Signatory



Signature:

Name: Siddharth Yadav

Date: 13th July 2009

7. Document References

- /1/ Registered CDM PDD V3.0 dated 14/11/2007.
- /2/ ACM0002: Consolidated methodology for grid-connected electricity generation from renewable sources, version 06, dated 19/05/2006.
- /3/ VER Monitoring Report of the Project:
Version 1, dated 25/09/2008;
Version 2, dated 08/05/2009.
- /4/ Validation report for the project issued by DNV, Report No. 2007-0698, Rev. No. 01, dated 21/12/2007.
- /5/ CDM registration information on UNFCCC website:
<http://cdm.unfccc.int/Projects/DB/DNV-CUK1198674466.99>.
- /6/ Calibration reports of the meters involved.
- /7/ Accreditation Certificates of Chifeng Electricity Bureau Electricity Measuring Institute and Northeast Electric Science Researching Institute.
- /8/ Daily Reading Records of the Project (from 01/01/2007 to 30/06/2008)
- /9/ Monthly Reading Records of the Project (from 01/01/2007 to 30/06/2008)
- /10/ ER Spreadsheet
- /11/ Operation Manual, Monitoring plan and staff training records.
- /12/ Project owner's statement on not included in an emission trading program.
- /13/ LoA from the Chinese DNA for the project.
- /14/ Voluntary Carbon Standard 2007.1 dated 18/11/2008.
- /15/ VCS Guidance Document: VCS Project Registration and VCU Issuance Process version 1.1.
- /16/ Further Guidance for Projects that are Registered in Two GHG Programs (19 March 2008, see VCS Website).

A.1 Annex 1: Findings Overview

Findings Overview Summary

	CARs	CLs	FARs
Total Number raised	0	1	0

Date:	23/02/2009	Raised by:	Simon Zhao Xinguang		
Type:	CL	Number:	CL #1	Reference:	Section 3
Lead Assessor Comment:			Date: 23/02/2009		
Please report all the calibrations carried out for all the meters involved to demonstrate that all meters were calibrated annually.					
Project Participant Response:			Date: 08/05/2009		
All these meters have been calibrated annually and evidences have been provided. The 10kv backup line meter was calibrated by Chifeng Electricity Bureau Electricity Measuring Institute in 05/2007 and 05/2006 and 05/2008. Other meters were calibrated by Northeast Electric Science Researching Institute. The meter measuring E _{Sunjiaying} and E _{Wudaogou} were calibrated in 11/2006, 05/2007and 05/2008 and the meter measuring E _{Export} and E _{import} was calibrated in 11/2006 and 10/2007. Both entities have relevant qualifications for meter calibration. Detailed calibration information is reported in version 2 of the monitoring report.					
Documentation Provided as Evidence by Project Participant:					
Calibration Records of the gateway meter; Calibration records of the meters of Wudaogou wind farm; Calibration records of the meters of Sunjiaying wind farm; Calibration records of the backup line meter; Qualification of Chifeng Electricity Bureau Electricity Measuring Institute and Northeast Electric Science Researching Institute. Monitoring Report, version 2, dated 08/05/2009.					
Information Verified by Lead Assessor:					
Calibration Records of the gateway meter Calibration records of the meters of Wudaogou wind farm Calibration records of the meters of Sunjiaying wind farm Calibration records of the backup line meter Qualification of Chifeng Electricity Bureau Electricity Measuring Institute and Northeast Electric Science Researching Institute. Monitoring Report, version 2, dated 08/05/2009.					
Reasoning for not Acceptance or Acceptance and Close Out:					
The calibration records of meters involved and accreditation certificates of the calibrating entities were verified. It is confirmed that the meters were calibrated annually as per the monitoring plan in the registered CDM PDD. CL #1 is closed out.					
Acceptance and Close out by Lead Assessor: Simon Zhao Xinguang			Date: 01/06/2009		

A.2 Annex 2: Competency statements of team members

Statement of Competence

Name: Simon Zhao Xinguang

SGS Affiliate: SGS China

Status

- Product Co-ordinator ☐
- Operations Co-ordinator ☐
- Technical Reviewer ☐
- Expert ☐

Validation Verification

- Local Assessor ☒ ☒
- Lead Assessor ☒ ☒
- Assessor ☒ ☒
- / Trainee Lead Assessor

Scopes of Expertise

1. Energy Industries (renewable / non-renewable) ☒
2. Energy Distribution ☐
3. Energy Demand ☒
4. Manufacturing ☐
5. Chemical Industry ☐
6. Construction ☐
7. Transport ☐
8. Mining/Mineral Production ☐
9. Metal Production ☐
10. Fugitive Emissions from Fuels (solid, oil and gas) ☐
11. Fugitive Emissions from Production and Consumption of Halocarbons and Sulphur Hexafluoride ☒
12. Solvent Use ☐
13. Waste Handling and Disposal ☐
14. Afforestation and Reforestation ☐
15. Agriculture ☐

Approved Member of Staff by Siddharth Yadav Date: 20/10/2008

Statement of Competence

Name: Ginger Jiang Yuan

SGS Affiliate: SGS China

Status

- Product Co-ordinator ☐
- Operations Co-ordinator ☐
- Technical Reviewer ☐
- Expert ☐

Validation

Verification

- Local Assessor ☒
- Lead Assessor ☒
- Assessor ☒
- / Trainee Lead Assessor

Scopes of Expertise

1. Energy Industries (renewable / non-renewable) ☒
2. Energy Distribution ☐
3. Energy Demand ☐
4. Manufacturing ☐
5. Chemical Industry ☐
6. Construction ☐
7. Transport ☐
8. Mining/Mineral Production ☐
9. Metal Production ☐
10. Fugitive Emissions from Fuels (solid, oil and gas) ☐
11. Fugitive Emissions from Production and Consumption of Halocarbons and Sulphur Hexafluoride ☒
12. Solvent Use ☐
13. Waste Handling and Disposal ☒
14. Afforestation and Reforestation ☐
15. Agriculture ☐

Approved Member of Staff by Joe Sun

Date: 25/04/2009

MONITORING REPORT

Inner Mongolia Wudaogou 50.25MW Wind Power Project

Monitoring period: Jan. 1st, 2007 to Jun 30th, 2008

Date: May. 8th, 2009

Version: 2

1. Introduction

This document reports the emission reductions generated by Inner Mongolia Wudaogou 50.25MW Wind Power Project (hereinafter referred as the project) which has been registered successfully as a CDM project on Jul 5th, 2008 with the CDM registration reference number of 1487 before which on Dec. 27th, 2006, however, the project had partially commenced with real, quantifiable, additional and permanent emissions reductions which can be considered as Voluntary Carbon Units (VCUs) under the Voluntary Carbon Standard 2007.1 (VCS 2007.1). The monitoring period is from Jan. 1st, 2007 to Jun. 30th, 2008.

In addition, this monitoring report also refers to the registered CDM PDD (Version 3.0 14/11/2007).

2. General description of the project

2.1 Project activity

Inner Mongolia Wudaogou 50.25MW Wind Power Project involves the installation of 67 sets of 750KW wind turbines, accumulating to a total installed capacity of 50.25MW. The project is estimated that the annual generation of the project will be 124.66GWh. As a result, annual 142,848 tones of CO₂ emission reduction will be achieved by replacing equivalent electricity generated by Northeast China Power Grid, which is dominated by thermal power plants.

The project can reduce GHG emissions by replacing the generation from fossil fuel fired power plants by which the Northeast China Power Grid is dominated. And also, it contributes to sustainable development in the region by reducing pollutants discharge (e.g. SO₂, NO_x and dust), creating employment opportunities, promoting the local economic development.

2.2 Technical description of the project

Location of the project activity

The project is located in Wengniute county, East Inner Mongolia, its geographical coordinates are north latitude 42°39'-42°42' and east longitude 117°54'-118°01', and its altitude is around 1700m. It is 90km away from Wudan town, where Wengniute county government is located.

Technology employed by the project activity

Totally 67 wind turbines with nominal capacity of 750 kW have been installed with 50m as the turbine installation height, providing a total capacity of 50.25 MW. Xinjiang Gold Wind 750kW (S48/750) wind turbines were adopted in the project. Each turbine has a 690V-to-35kV transformer, from which a 35kV line links into the on-site 220kV switchgear at the

substations established in the project site. By the 220 kV line, the electricity generated by the project are delivered to the power grid.

The first wind turbine was put into operation on 27th Dec. 2006 and the last wind turbine was put into operation on 31st Dec. 2007.

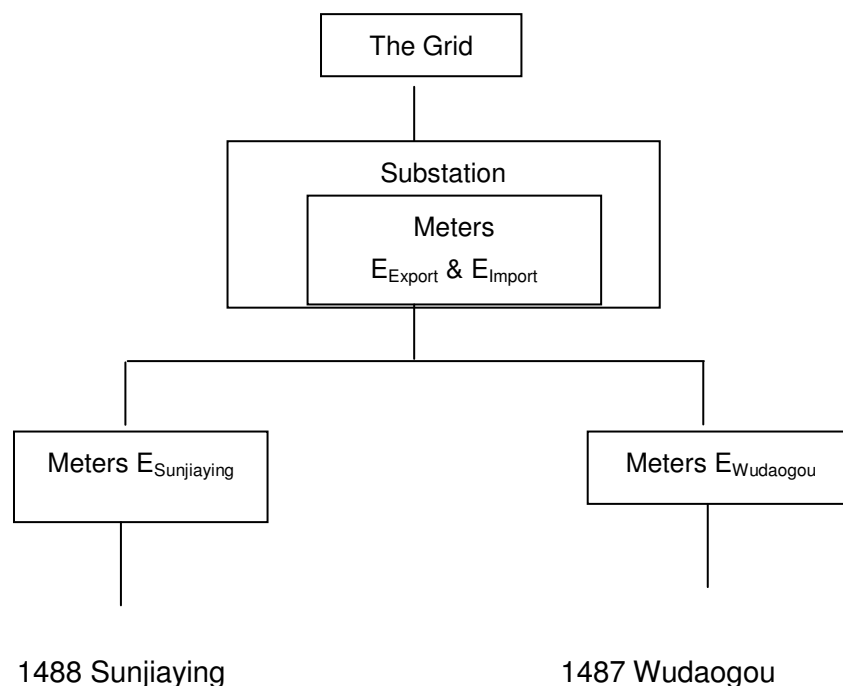
3. Monitoring methodology and plan

The monitoring methodology in this report refers to the CDM monitoring methodology ACM0002 (ver.06) – “Consolidated monitoring methodology for grid-connected electricity generation from renewable sources”.

Inner Mongolia Sunjiaying 50.25MW Wind Power Project and Inner Mongolia Wudaogou 50.25MW Wind Power Project are owned by the same project entity. The two wind farms are sharing the power line. Total net electricity delivered to the grid by the two wind farms can be measured by one shared meter. This shared meter has bidirectional metering, recording both export to the grid (E_{Export}) and import from the grid (E_{Import}).

Separate meters are installed at each of the project sites, which can also record bidirectional metering, and can be used as back-up for the main meter. The metering equipment at the project site are owned, operated and maintained by Chifeng Xinsheng Wind Power Co. Ltd.

A metering diagram is presented below, giving an overview of these meters.



The net electricity delivered to the grid from each of wind farms can be calculated by the

readings of the shared meter and the meters measuring power generation from each wind farm. The measurement of electricity from Sunjiaying and Wudaogou are used to calculate the share of each project in the supply to the grid. The project activity's share in imports from the grid is not separately calculated, in order to ensure the conservativeness.

The grid company provides the result to the project owner monthly and the project owner checks the results against the readings from the meters directly measuring the two projects.

According to the method determining the electricity imported and exported by the two projects between the project owner and the grid company, readings of the shared meter and the project-wise meters are used to calculate the share of net electricity delivered to the grid by each project. The calculation formula shows as follows:

$$EG_y = E_{Export} / (E_{Sunjiaying} + E_{Wudaogou}) * E_{Wudaogou} - E_{import} - E_{10kv}$$

Where:

EG_y is the calculated net electricity supplied from the project activity;

E_{Export} is the total electricity supplied to the grid, this data is provided by the grid company and used in the Electricity Transaction Note, and it is checked by the meter reading from the onsite substation (which is for the two wind farms combined);

$E_{Sunjiaying}$ is the electricity generation metered by the onsite meters of 1488 Inner Mongolia Sunjiaying 50.25MW Wind Power Project;

$E_{Wudaogou}$ is the electricity generation metered by the onsite meters of 1487 Inner Mongolia Wudaogou 50.25MW Wind Power Project;

E_{Import} is the total electricity imported from the grid by the two projects; this data is measured by the grid company and confirmed by electricity purchase invoice

E_{10kv} is the electricity consumption through the 10kV back-up power line by the project activity, which is also confirmed by the grid company.

Table 1 Monitored data and determination of EG_y

Period	E_{Export} (kWh)	E_{Import} (kWh)	E_{10kv} (kWh)	$E_{Sunjiaying}$ (kWh)	$E_{Wudaogou}$ (kWh)	EG_y (kWh)
	A	B	C	D	E	$F=A/(D+E)*E-B-C$
01/01/07- 31/01/07	23700000	94380	4860	11398800	12094250	12101548
01/02/07- 28/02/07	28640000	26180	0	14074550	15089900	14792366
01/03/07- 31/03/07	19040000	125840	0	9941750	10912300	9837222
01/04/07- 30/04/07	22820000	43340	0	11558750	10221750	10666255
01/05/07- 31/05/07	24346740	31900	60	10818150	9370200	11268311
01/06/07- 30/06/07	6420000	91960	0	4807950	4927300	3157393

01/07/07-31/07/07	4490000	124960	600	2065700	2354450	2266097
01/08/07-31/08/07	9301600	109340	60	3761450	2727200	3800092
01/09/07-30/09/07	6040000	105380	16440	3138450	3177650	2916923
01/10/07-31/10/07	26640000	96800	14040	14788200	12820500	12259830
01/11/07-30/11/07	41250000	65340	6960	26036150	16751000	16076912
01/12/07-31/12/07	46161940	32340	9540	28031150	18452350	18282822
<i>Subtotal 2007</i>						<i>117425771</i>
01/01/08-31/01/08	45500000	73480	5520	29636950	19304250	17867911
01/02/08-29/02/08	48460000	44440	3840	30705850	19533150	18793187
01/03/08-31/03/08	37510000	62040	4600	28035000	16051700	13590523
01/04/08-30/04/08	37490000	67540	31540	25352950	13493200	12923502
01/05/08-31/05/08	35751980	66440	18300	23843400	11959850	11857984
01/06/08-30/06/08	9035400	121220	9720	5711650	3442600	3266965
<i>Subtotal 2008</i>						<i>78300072</i>
Total	472597660	1382480	126080	283706850	202683600	195725841

Total electricity supplied by the project in 2007 was 117425771 kWh and in 2008 was 78300072 kWh.

4. Quality Control (QC) and Quality Assurance (QA)

4.1 QA/QC procedures

ID	Uncertainty level of data (High/Medium/Low)	Explain QA/QC procedures planned for these data, or why such procedures are not necessary.
EG _y	Low	The net electricity used for the calculation of the emission reduction is obtained through the corresponding share of total exported generation subtracts the imported electricity. The electricity is monitored accurately by the main meter (bidirectional function) installed at the exit side of the 220kV transformer station near the wind farm. Electricity generated by the project and that by Sunjiaying project are measured continuously and recorded every day by the project owner. In the

		meanwhile, electricity consumed by the two projects are measured and recorded by the project owner. The total electricity export is measured and recorded by both the project owner and the grid company. Furthermore, electricity sales invoices and electricity transaction notes have been provided for double check. Electricity imported from the 10kv backup line is measured by the grid company and recorded monthly. Electricity sales invoices have been made available to the verifier. Meters are calibrated at least once a year. Calibration certificates of meters have been made available to the verifier.
--	--	--

Meters involved for monitoring

Data measured	Manufacturer	Type	Serial number	Accuracy	Calibration date
$E_{\text{Export}} \& E_{\text{Import}}$	Landisgyr	ZD402	86693424	0.2S	2007.10.25 2006.11.09
E_{wudaogou}	Changsha Weisheng	DTSD341	200504230U0190	0.5S	2008.05.19 2007.05.19 2006.11.16
E_{wudaogou}	Changsha Weisheng	DTSD341	200504230U0242	0.5S	2008.05.19 2007.05.19 2006.11.16
$E_{\text{sunjiaying}}$	Changsha Weisheng	DTSD341	200504230U0186	0.5S	2008.05.19 2007.05.19 2006.11.16
$E_{\text{sunjiaying}}$	Changsha Weisheng	DTSD341	200504230U0245	0.5S	2008.05.19 2007.05.19 2006.11.16
$E_{10\text{kv}}$	Zhejiang Zhengtai	DTS634	2006-04-46355251	1.0S	2008.05.19 2007.05.28 2006.05.28

All meters have been calibrated once a year according to the registered PDD.

4.2 Roles and responsibilities

The Monitoring Plan clearly states the persons and the responsibilities of the QC team of the project; they are responsible for monitoring the electricity generation of the project.

5. GHG Calculations

According to the registered PDD: $ER_y = BE_y - PE_y - Ly$

5.1 Project activity emissions

According to the registered PDD, the project activity does not have any GHG emissions.

$$PE_y=0$$

5.2 Leakages

According to the registered PDD, the leakage from the proposed project is zero.

$$L_y=0$$

5.3 Baseline emissions

The baseline emission BE_y during the monitoring period results from:

$$BE_y = EG_y \times EF_y$$

Where:

EG_y – the net power transmitted to the grid during the monitoring period year y (kWh);

EF_y – the emission factor of the grid in the year y (calculated ex-ante).

The net power transmitted to the grid (EG_y) was 117425.6 MWh in 2007 and 78300MWh in 2008.

The emission factor of the proposed project (EF_y) is 1.1459tCO₂e/MWh¹ (the weights of OM and BM are 0.75 and 0.25, respectively).

The baseline emission (BE_y) can be calculated by the formula below:

In year 2007

$$BE_y = EG_y \times EF_y = 117425.7 \text{ MWh} \times 1.1459 \text{ tCO}_2\text{e/MWh} = 134558 \text{ tCO}_2\text{e}$$

In year 2008

$$BE_y = EG_y \times EF_y = 78300 \text{ MWh} \times 1.1459 \text{ tCO}_2\text{e/MWh} = 89724 \text{ tCO}_2\text{e}$$

¹ Source: the final PDD for registration of the proposed project.

5.4 Emission reductions

As the emissions and leakage from the project activity are zero, emission reduction is equal to baseline emission.

Emission reductions were 134558tCO₂e in year 2007 and 89724tCO₂e in 2008.

6. Name of responsible persons

David WANG

United Carbon Assets Limited (The entity serves as a consultant and broker)

Room No.3-402 Tower 1,

Jinquan Square, Datun Road,

Chaoyang District, Beijing 100101

Tel./Fax: +86 -10-51501100/ +86-10-51501050

E-mail: David@ucahk.com

Annex 1

Clauses 1.12, 1.13, 1.14, 8.1 and 8.2 of the VCS Project

Description Template

Clause 1.12: Demonstration to confirm that the project was not implemented to create GHG emissions primarily for the purpose of its subsequent removal or destruction

The project is a renewable resource based power project, which discharges no emission when operating at all. The project was implemented primarily to create electricity from wind energy and its main revenue is from the sale of electricity. Therefore, it can be confirmed that the project was not implemented to create GHG emissions for its subsequent removal or destruction.

Clause 1.13: Demonstration that the project has not created another form of environmental credit (for example renewable energy certificates).

The project owner declares that the project has not created any other form of environmental credit, which will be verified by DOE.

Clause 1.14: Project rejected under other GHG programs (if applicable).

The project has been registered as a CDM project on Jul 5th, 2008, for which a renewable crediting period of 3×7 years will be used under the CDM GHG Program. Therefore, CO2 emission reductions generated by the project during the CDM crediting period which start from 5th Jul. 2008 will be verified as unique CERs but not VCUs to avoid double counting.

Clause 8.1: Proof of Title.

The project including the plant, equipment and its emissions reductions is owned by Chifeng Xinsheng Wind Power Co., Ltd. Relevant evidences for the ownership have been provide to the DOE, which involves business license of the project company issued by Wengniute County Industrial and Commercial Administrative Bureau, and Letter of Approval for the project as a CDM project by Chinese NDRC.

Clause 8.2: Projects that reduce GHG emissions from activities that participate in an emissions trading program (if applicable).

The project takes place in China, which has no emission reductions target and the project is not a micro or mega project. There are no mandatory emission trading programmes in China. The project is not a grouped one and the project is a registered CDM project activity and not rejected under other GHG programs. The ERs generated by the project in the monitoring period will only to be sold once and reported to VCS registry, therefore impossible to sell again.

A.4 Annex 4: Revised Monitoring Plan

B.7.1 Data and parameters monitored:

Data / Parameter:	E_{Export}
Data unit:	MWh
Description:	Electricity delivered to the grid by Wudaogou and Sunjiaying projects
Source of data to be used:	Directly measured
Value of data applied for the purpose of calculating expected emission reductions in section B.5	N/A
Description of measurement methods and procedures to be applied:	The metering equipment will be properly calibrated annually according to the requirement from Technical administrative code of electric energy metering (DL/T448 - 2000). The meter is installed at the transformer station, which measures the total electricity supplied to the grid. Through the electronic multifunctional electricity meter (accuracy degree is 0.2S, bidirectional), the electricity supplied to the grid is measured. The readings of electricity meter will be continuously measured and monthly recorded. Data will be archived for 2 years following the end of the crediting period by means of electronic and paper backup.
QA/QC procedures to be applied:	Electricity sales invoices will also be obtained for double check.
Any comment:	Electricity supplied by the two project activities to the grid. Double check by receipt of sales.

Data / Parameter:	E_{Import}
Data unit:	MWh
Description:	Electricity imported from the grid by Wudaogou and Sunjiaying projects
Source of data to be used:	Directly measured
Value of data applied for the purpose of calculating expected emission reductions in section B.5	0
Description of measurement methods and procedures to be applied:	The metering equipment will be properly calibrated annually according to the requirement from Technical administrative code of electric energy metering (DL/T448 - 2000). The meter is installed at the transformer station,

	which measures the total electricity supplied to the grid. Through the electronic multifunctional electricity meter (accuracy degree is 0.2S, bidirectional), the electricity supplied to the grid is measured. The readings of electricity meter will be continuously measured and monthly recorded. Data will be archived for 2 years following the end of the crediting period by means of electronic and paper backup.
QA/QC procedures to be applied:	Electricity sales invoices will also be obtained for double check.
Any comment:	Electricity consumed by the two project activities to the grid. Double check by electricity purchase invoices.

Data / Parameter:	E_{Wudaogou}
Data unit:	MWh
Description:	Electricity generated by Wudaogou project
Source of data to be used:	Directly measured
Value of data applied for the purpose of calculating expected emission reductions in section B.5	124660
Description of measurement methods and procedures to be applied:	The metering equipment will be properly calibrated annually according to the requirement from Technical administrative code of electric energy metering (DL/T448 - 2000). The meter is installed at the transformer station, which measures the electricity generation of Wudaogou project. The meter is electronic multifunctional with an accuracy degree of 0.5S. The readings of electricity meter will be continuously measured and monthly recorded. Data will be archived for 2 years following the end of the crediting period by means of electronic and paper backup.
QA/QC procedures to be applied:	The electricity output from each turbine will be monitored and recorded.
Any comment:	

Data / Parameter:	E_{Sunjiaying}
Data unit:	MWh
Description:	Electricity generated by Sunjiaying project
Source of data to be used:	Directly measured
Value of data applied for the purpose of calculating expected emission reductions in section B.5	120940
Description of measurement methods and procedures to be applied:	The metering equipment will be properly calibrated annually according to the requirement from Technical administrative code of electric energy

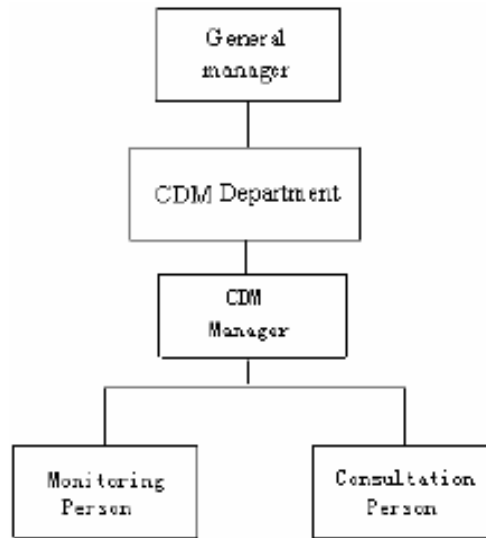
	metering (DL/T448 - 2000). The meter is installed at the transformer station, which measures the electricity generation of Sunjiaying project. The meter is electronic multifunctional with an accuracy degree of 0.5S. The readings of electricity meter will be continuously measured and monthly recorded. Data will be archived for 2 years following the end of the crediting period by means of electronic and paper backup.
QA/QC procedures to be applied:	The electricity output from each turbine will be monitored and recorded.
Any comment:	

Data / Parameter:	E_{10kv}
Data unit:	MWh
Description:	Electricity imported from 10kv backup line
Source of data to be used:	Directly measured
Value of data applied for the purpose of calculating expected emission reductions in section B.5	0
Description of measurement methods and procedures to be applied:	The metering equipment will be properly calibrated annually according to the requirement from Technical administrative code of electric energy metering (DL/T448 - 2000). The meter is maintained and recorded by the grid company, and its accuracy degree of 1.0. Data will be archived for 2 years following the end of the crediting period by means of electronic and paper backup.
QA/QC procedures to be applied:	Electricity purchase invoices will also be obtained for double check.
Any comment:	

Data / Parameter:	EG_y
Data unit:	MWh
Description:	Electricity supplied by Wudaogou project
Source of data to be used:	Calculated
Value of data applied for the purpose of calculating expected emission reductions in section B.5	N/A
Description of measurement methods and procedures to be applied:	EG_y will be calculated as follows: $EG_y = E_{\text{Export}} / (E_{\text{Sunjiaying}} + E_{\text{Wudaogou}}) * E_{\text{Wudaogou}} - E_{\text{Import}} - E_{10kv}.$
QA/QC procedures to be applied:	N/A
Any comment:	N/A

B.7.2. Description of the monitoring plan:

This monitoring plan will be implemented by professional staff authorized by the owner of the proposed project. The management structure is illustrated as follows:



Annex 4

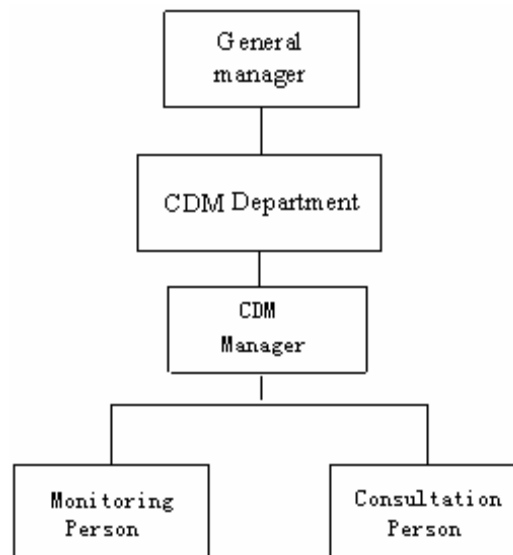
MONITORING INFORMATION

Monitoring plan is a division and schedule of a series of monitoring tasks. Monitoring tasks must be implemented according to the monitoring plan in order to ensure that the real, measurable and long-term greenhouse gas (GHG) emission reduction for the proposed project is monitored and reported.

1. Management structure and staff for implementation of monitoring plan

The Management Group of the proposed project must maintain credible, transparent, and adequate data estimation, measurement, collection, and tracking systems to maintain the information required for an audit of an emission reduction project.

The Management Group consists of professional staff authorized by the owner of the proposed project, i.e. Chifeng Xinsheng Wind Power Co. Ltd. The management structure is illustrated as follows:



The Management Group is responsible for the CDM management of the proposed project, which specifically includes selecting a CDM consultant company, providing support to the selected consultant company, assisting the selected consultant company in selecting CER buyer and determine CER price, assisting the selected consultant company in applying for China DNA approval, providing support in the due diligence of the selected buyer, providing support in the CDM validation and registration by a selected DOE, and providing support in the verification by a selected DOE.

The General Manager is in general control and makes key decisions. The CDM Department is responsible for specific tasks in monitoring and execution of decisions. Specifically, the CDM Manager is responsible for the daily operation of the CDM Department, and contact with DOE and EB to support their work in validation, registration, verification, and certification. The procedure is illustrated in Section 5 of this monitoring plan. He is also responsible for Monitoring Data adjustment and settlement of Data uncertainties, together with local electric

power company, and/or DOE. The procedure is illustrated in Section 3 of this monitoring plan.

The Monitoring Person is responsible for reading and calibration of the meter, recording of the readings, and reporting of readings to local electric power company and/or DOE. The procedures for reading, reporting, and calibration are illustrated in Section 2 and Section 4 of this monitoring plan, respectively.

The Consultation Person is responsible for selecting a CDM consultant company and providing support to its work in developing PDD, selecting CER buyer, and applying for DNA approval.

The Management Group has all received sufficient training in terms of monitoring and verification. They have received general training on wind power project operation organized by the project owner, including reading and calibration of the meter, recording of the readings, adjustment of readings, and reporting of readings. On the other hand, they have received CDM training organized by China Fulin Windpower Development Corporation, including validation, registration and verification. When necessary, the CDM Manager is responsible for organizing or attending trainings on Monitoring and Verification. The procedure is as follows:

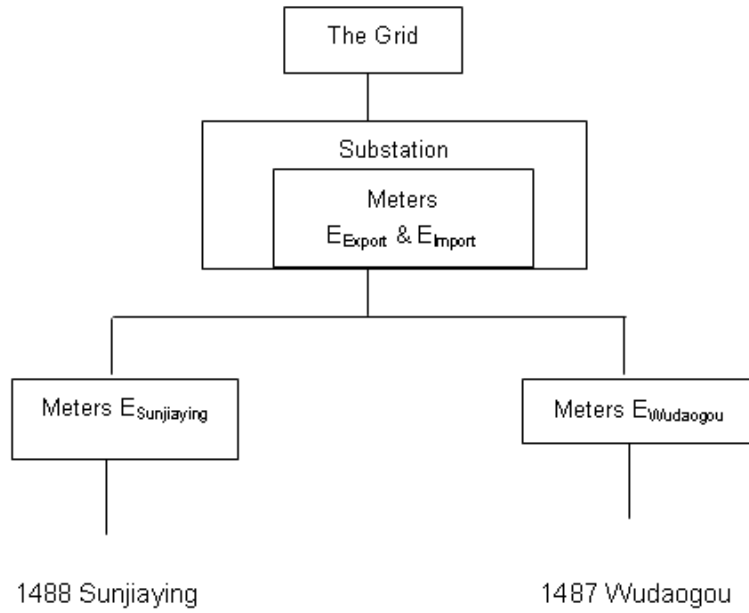
- (1) Investigating whether there is need for trainings, and if so, the content of trainings.
- (2) If so, obtaining the approval of General Manager.
- (3) Checking if there are any trainings on Monitoring and Verification to be organized by other organizations. If so, attending them.
- (4) If not, organizing trainings themselves.

2. Data and Approach for Monitoring

Inner Mongolia Sunjiaiyng 50.25MW Wind Power Project is located near to Inner Mongolia Wudaogou 50.25MW Wind Power Project; therefore the grid company refused to build two power lines for the two projects and instructed them to share one power line. Total net electricity delivered to the grid by the two wind farms can be measured by one shared meter. Thus the net delivery of Sunjiaying project cannot be directly measured and have to be calculated according to records of several meters. The shared meter has bidirectional metering, recording both export to the grid (E_{Export}) and import from the grid (E_{Import}).

Separate meters are installed at each of the project sites. The project-wise metering equipment at the project site are owned, operated and maintained by Chifeng Xinsheng Wind Power Co. Ltd.

A metering diagram is presented below, giving an overview of these meters.



The net electricity delivered to the grid from each of wind farms can be calculated by the readings of the shared meter and the project-wise meters at the substation. The project-wise metering equipment are used to calculate the share of each of the projects of the net supply to the grid. The project activity's share in exports to the grid and imports from the grid is not separately calculated, as only net supply is required for the emission reduction calculations. The grid company provides the result to the project owner monthly and the project owner checks the results against the readings from the project-wise meters.

According to the method determining the electricity imported and exported by the two projects between the project owner and the grid company, readings of the shared meter and the project-wise meters are used to calculate the share of net electricity delivered to the grid by each project. The calculation formula shows as follows:

$$EG_y = E_{Export} / (E_{Sunjiaying} + E_{Wudaogou}) * E_{Wudaogou} - E_{import} - E_{10kv}$$

Where:

EG_y is the calculated net electricity supplied from the project activity;

EG_{Export} is the total net electricity supplied to the grid, i.e. exports minus imports, this data is provided by the grid company and used in the Electricity Transaction Note, and it is checked by the meter reading from the onsite substation (which is for the two wind farms combined);

$E_{Sunjiaying}$ is the electricity generation metered by the onsite meters of 1488 Inner Mongolia Sunjiaying 50.25MW Wind Power Project;

$E_{Wudaogou}$ is the electricity generation metered by the onsite meters of 1487 Inner Mongolia Wudaogou 50.25MW Wind Power Project;

E_{Import} is the total electricity imported from the grid by the two projects; this data is measured by the grid company and confirmed by electricity purchase invoices

E_{10kv} is the electricity consumption through the 10kV back-up power line by the project activity, which is also confirmed by the grid company.

The Data To Be Monitored :

ID number	Data variable	Source of data	Data unit	Measured (m), calculated (c), estimated (e),	Recording frequency	Portion of data to be monitored	How will the data be archived? (electronic /paper)	For how long is archived data kept?	Comment
1. E_{Export}	Electricity supplied to the grid by the two projects	meter	MWh	m	continuously measured and monthly recorded	100%	electronic	During the crediting period and two years after	
2. $E_{Wudaogou}$	Electricity generation of Wudaogou project	meter	MWh	m	continuously measured and monthly recorded	100%	electronic	During the crediting period and two years after	
3. $E_{Sunjiaying}$	Electricity generation of Sunjiaying project	meter	MWh	m	continuously measured and monthly recorded	100%	electronic	During the crediting period and two years after	
4. E_{Import}	Electricity imported from the grid	meter	MWh	m	continuously measured and monthly recorded	100%	electronic	During the crediting period and two years after	
5. E_{10kv}	Electricity from the 10kv backupline	meter	MWh	m	continuously measured and monthly recorded	100%	electronic	During the crediting period and two years after	
6. E_{Gy}	Electricity supplied to the grid by Wudaogou project	meter	MWh	c	Monthly calculated and recorded	100%	electronic	During the crediting period and two years after	

The specific steps to monitoring are listed below:

- The project owner reads the meter and records data on the last day of every month
- The project owner provides electricity sales invoice to Liaoning Electric Power Company.
- The project owner provides the meter's data readings to DOE for verification.

The meter reading will be readily accessible for DOE. Calibration test records will be maintained for verification.

3. Calibration of Meters & Metering

The metering equipments have been properly calibrated annually according to the requirement from Technical administrative code of electric energy metering (DL/T448 - 2000). The metering equipments were checked by the project owner and Liaoning Electric Power Company before operation.

☐ If any of the below circumstances happened on any meter, the meter shall be tested by the qualified metrical organization co-authorized by the Northeast Power Grid and the project owner:

- 1) The detection of a difference larger than the allowable error in the reading of the meter, when considering the reactive loss of electrical wire,
- 2) The repair of all or part of the meter caused by the failure of one or more parts to operate in accordance with the specifications,
- 3) If any errors are detected, the party owning the meter shall be repaired, recalibrate or replace the meter and give the other party sufficient notice to allow a representative to attend during any corrective activity.

☐ Should reading of the meter be inaccurate by more than the allowable error, or otherwise functioned improperly, the electricity supplied to the grid by the proposed project shall be determined by:

- 1) First, by reading the self-carried meters of wind turbines, unless a test by either party reveals they are inaccurate;
- 2) if the self-carried meters of wind turbines are not with acceptable limits of accuracy or are performing improperly, the project owner and the Liaoning Electric Power Company shall jointly prepare an estimate of the correct reading; and
- 3) If the project owner and the Liaoning Electric Power Company fail to agree the estimate of the correct reading, then the matter will be referred for arbitration according to agreed procedures.

☐ The electricity recorded by the meter will suffice for the purpose of billing and emission reduction verification as long as the error in the meter is within the permissible limits. Calibration is carried out by the Liaoning Electric Power Company with the records being provided to the project owner, and these records will be maintained by the project owner.

4. Data Management System

This part provides information on record keeping of the data collected during monitoring. Record keeping is the most important exercise in relation to the monitoring process. Below follows an outline of how project related records were managed.

The CDM manual sets out the procedures for tracking information from the primary source to data calculations, in paper format. If data and information are from internet, the website must be provided. Moreover, the credibility and reliability of those data and information from internet must be confirmed.

Physical documentation will be collated in a central place. In order to facilitate auditor's reference, monitoring results will be indexed. All paper-based information will be stored by the project owner. The following table below outlines the key documents relevant to monitoring and verification.

Table List of the key documents relevant to monitoring and verification

ID No.	Document Title	Main Content	Source
F-1	PDD, including the electronic spreadsheets and supporting documentation (assumptions, estimations, measurement, etc)	Calculation procedure of emission reduction and monitoring items	The project owner
F-2	The report on monitoring and checking of electricity supplied to the grid	Record based on monthly meter reading and electricity sale receipts	The project owner
F-3	Report on maintenance and calibration of metering equipment	Reasons for maintenance and calibration and the precision after maintenance and calibration	The project owner

5. Verification and Monitoring Results

The verification of the monitoring results of the project is a mandatory process required for all CDM projects. The main objective of the verification is to independently verify that the project has achieved the emission reductions as reported and projected in the PDD. It is expected that the verification will be done annually.

The responsibilities for verification of the projects are as follows:

- ◆ Sign a verification service agreement with specific DOE and agree to a time framework set by the EB for carrying out verification activities while taking into account the buyer's schedule. The project owner will make the arrangements for the verification and will prepare for the audit and verification process to the best of its abilities.
- ◆ The project owner will facilitate the verification through providing the DOE with all required necessary information, before, during and, in the event of queries, after the verification.
- ◆ The project owner will fully cooperate with the DOE and instruct its staff and management to be available for interviews and respond honestly to all questions from the DOE.
- ◆ DOE must be an Accredited Entity with a proven track record in environmental auditing and verification, experience with CDM projects and work in developing countries. The DOE

should be accredited by the CDM EB. If the project owner deems that requirements of DOE go beyond the scope of verification, they should determine whether the requirements of DOE are reasonable. If considered unreasonable, a rejection letter in a written format should be provided to the DOE with justifiable reasons. If the project owner and the DOE cannot reach an agreement, the matter will be submitted to EB or UNFCCC for arbitration.